



Linear Control Systems 25411

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Assignment 4

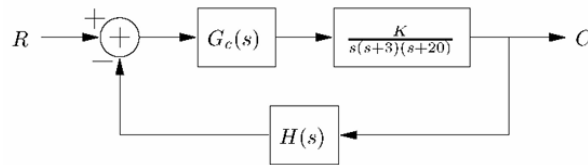
Fall 1404

Due Date: 1404/9/28

Note 1: *purple* problems are bonus ones.

1 Root Locus Under A Microscope

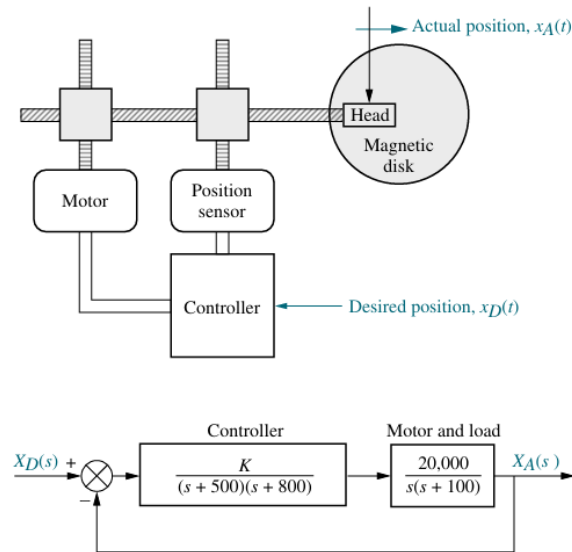
This problem will explore some of the nuances of root locus in systems with non-unity feedback.



- As a matter of familiarizing you with the system shown above, set $G_c(s) = 1$ and $H(s) = 1$. Plot the root locus of the loop transfer function $L(s)$, and specify the range of positive K for which the system is stable.
- Since this system is stable for an unacceptably small range of K , your friend Amir suggests one method of compensation; you follow his suggestion and set $G_c(s) = (s + 4)$ while leaving $H(s) = 1$. Plot the root locus of $L(s)$ with this compensator in place, and plot the locations of the closed-loop poles and zeros for $K = 200$. Additionally, use the step command in MATLAB to plot the step response of the closed-loop system with K still fixed at 200.
- As you are boasting about Amir's clever method of compensation, your other friend Mamali mentions that he has thought of another, more attractive, way of compensating the system. He suggests setting $G_c(s) = 1$, but making $H(s) = (s + 4)$. You are intrigued, and decide to compare Mamali's scheme to Amir's configuration. Plot the root locus of $L(s)$ with this compensator in place, and plot the locations of the closed-loop poles and zeros for $K = 200$. Plot the step response of this system in MATLAB as well with the same value of K as you were previously using.
- Are the step responses different? If so, provide a short explanation for why this might be so.

2 Disk Drive Analysis

A disk drive is a position control system in which a read/write head is positioned over a magnetic disk. A physical representation of the system and a block diagram are shown in Figure below.



- Find K to yield a settling time of 0.1 second.
- What is resulting percent overshoot?
- What is the range of K that keeps the system stable?

3 Root Locus Shape

Show that for $0 < K < 4$, the root locus of the given open-loop transfer function, lies on a circle.

$$G(s) = \frac{s+2}{(s+1)^2}$$

4 Full Range Root Locus

The transfer function of a open-loop control system is given below:

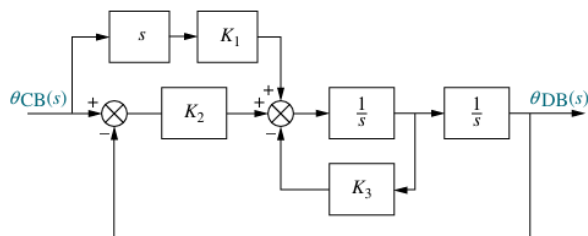
$$G(s) = \frac{s+3}{(s+5)(s-2)(s^2+2)}$$

Sketch the root locus plot as K is varied from $-\infty$ to $+\infty$ and determine:

- Angle of asymptotes and the centroid point
- The breaking-away points (if any)
- The range of K such the closed-loop system becomes unstable
- Angle of departure (from complex poles) and angle of arrival (to complex zeros) (if any)
- The value of K such that the closed-loop system is critically damped (if any)

5 Smoothing Procedure in Aerospace Applications

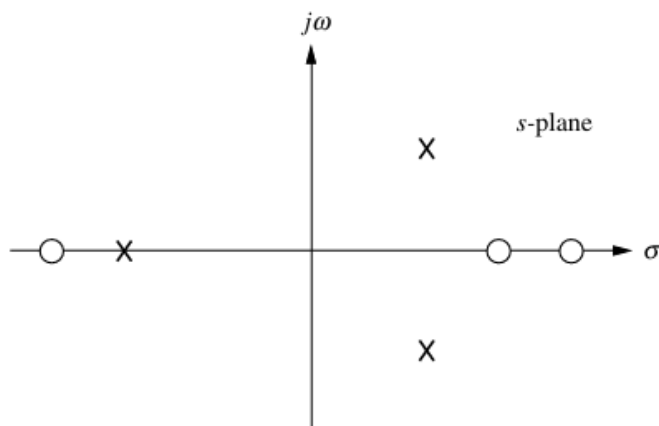
During ascent, the automatic steering program aboard the space shuttle provides the interface between the low-rate processing of guidance and the high-rate processing of flight control. The function performed is basically that of smoothing. A simplified representation of a maneuver smoother linearized for coplanar maneuvers is shown below. Here $\theta_{CB}(s)$ is the commanded body angle as calculated by guidance, and $\theta_{DB}(s)$ is the desired body angle sent to flight control after smoothing.



- Sketch a root locus where the roots vary as a function of K_3 .
- Locate the closed-loop zeros.
- Repeat Parts a and b for a root locus sketched as a function of K_2 .

6 Root Contours

Figure below shows open-loop poles and zeros. There are two possibilities for the sketch of the root locus. Sketch each of the two possibilities. Be aware that only one can be the real locus for specific open-loop pole and zero values.



7 Effect of Noise and Uncertainty

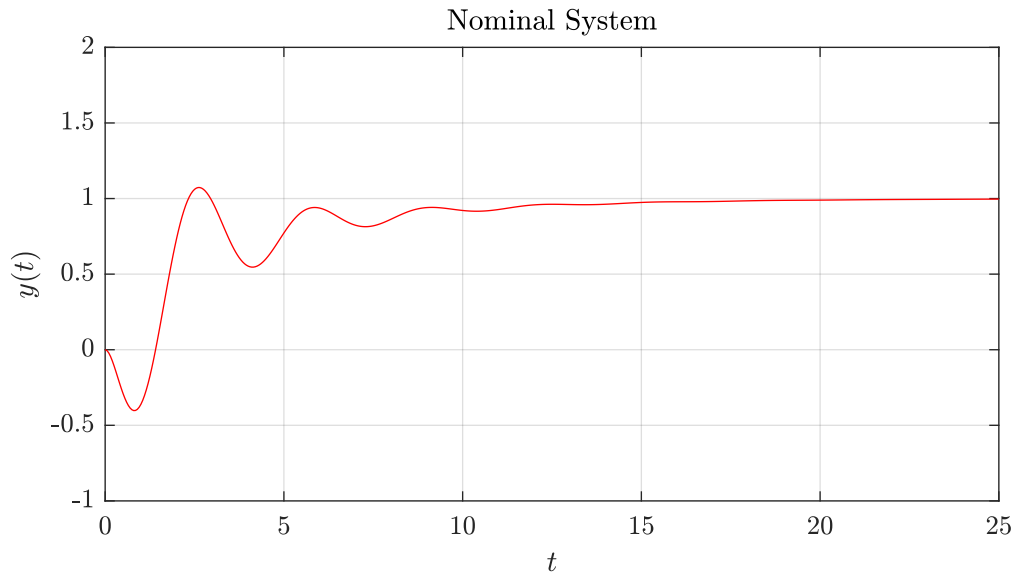
Consider the transfer function below:

$$G(s) = \frac{k(b_1s + 1)(1 - b_2s)\omega_n^2}{(\tau_1s + 1)(\tau_2s + 1)(s^2 + 2\zeta\omega_ns + \omega_n^2)}$$

where the nominal values of the system parameters are given as follows:

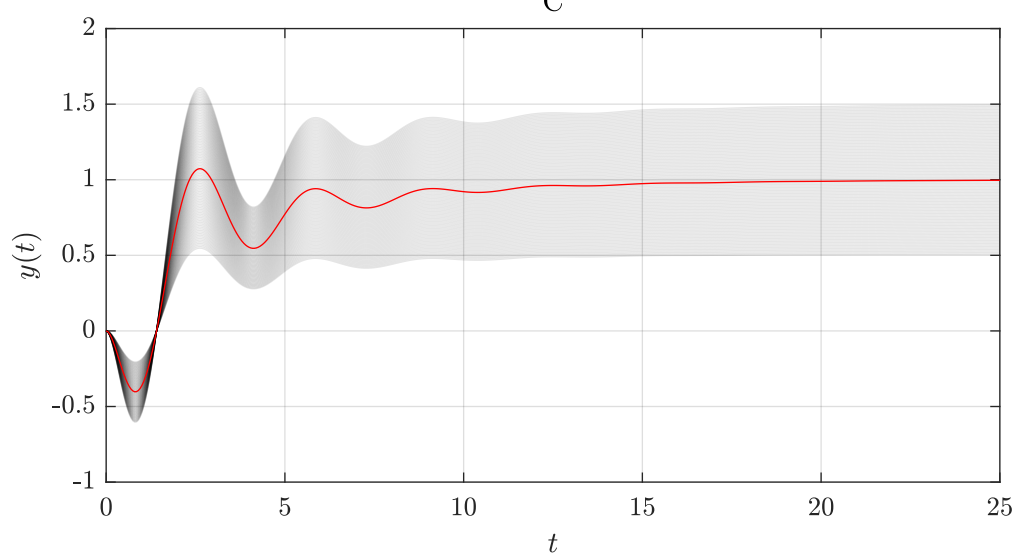
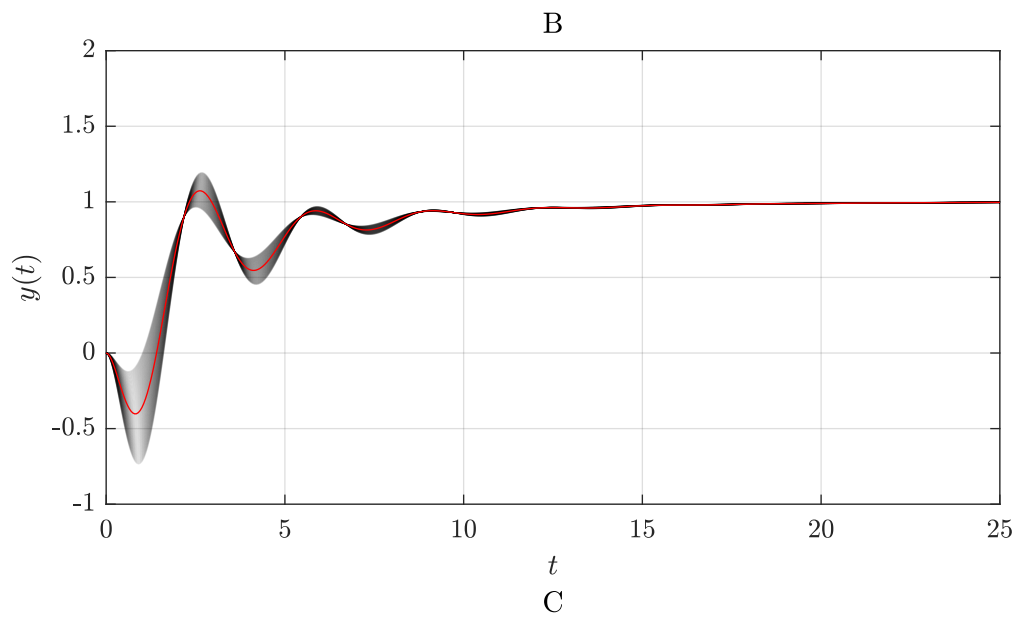
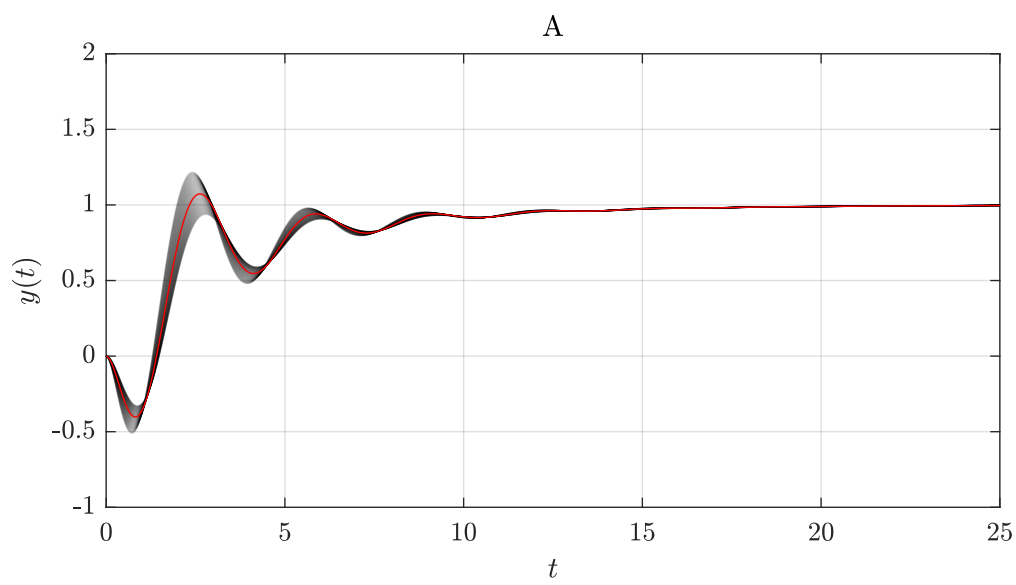
$\hat{k}$	1
$\hat{\tau}_1$	0.5
$\hat{\tau}_2$	5
$\hat{\zeta}$	0.2
$\hat{\omega}_n$	2
$\hat{b}_1$	3
$\hat{b}_2$	1

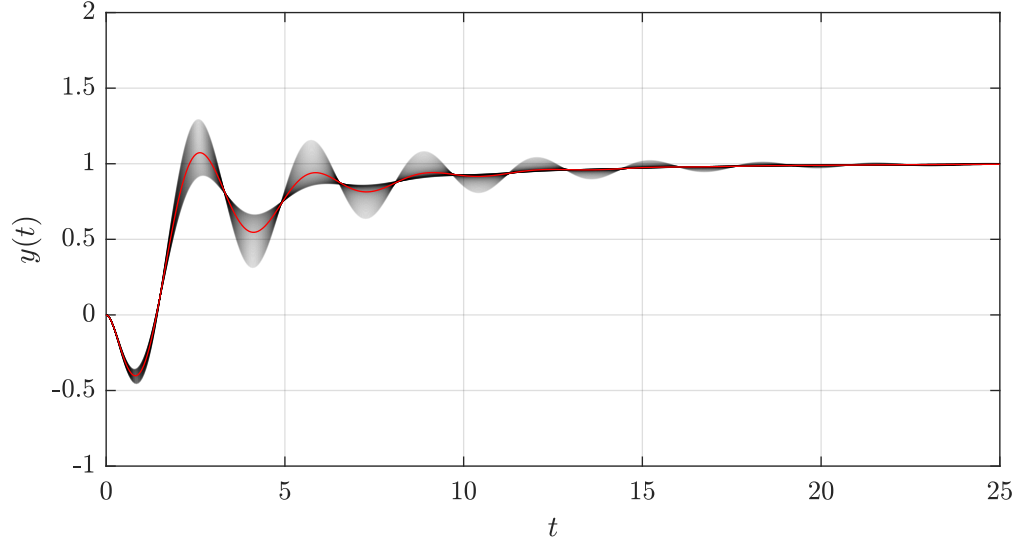
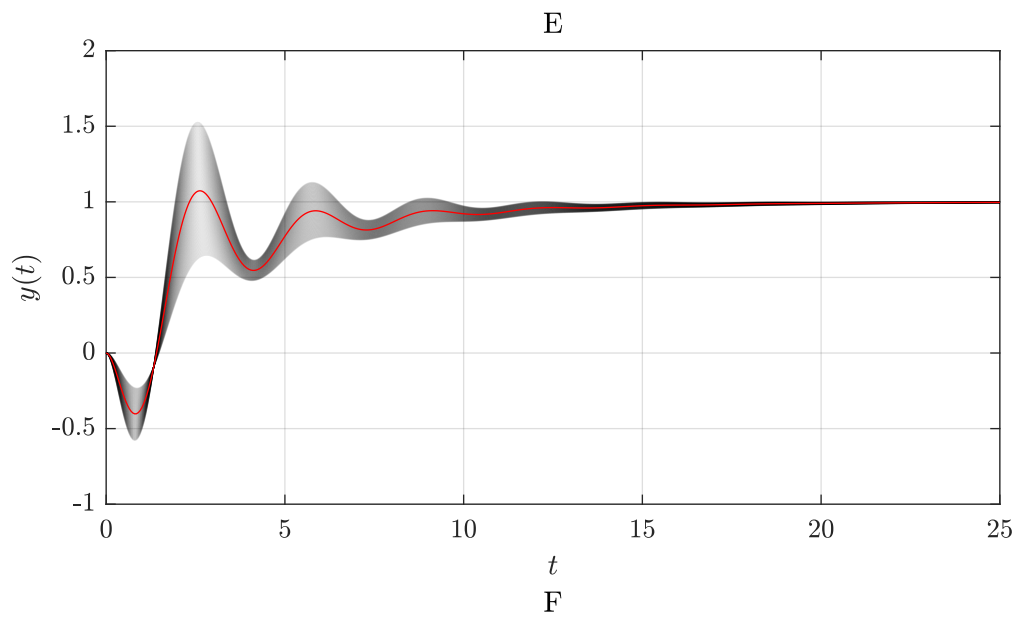
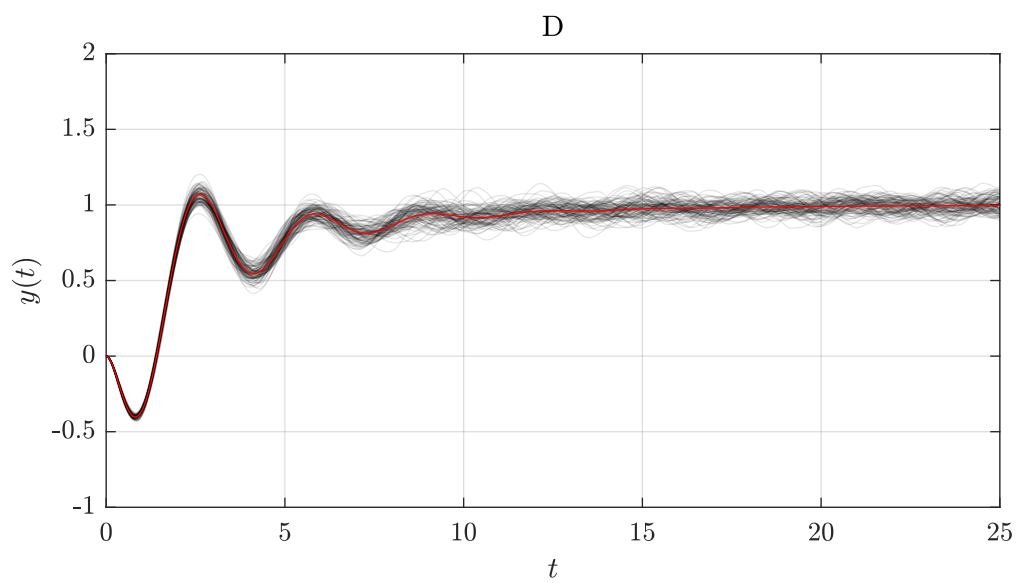
The nominal step response of system is as follows:

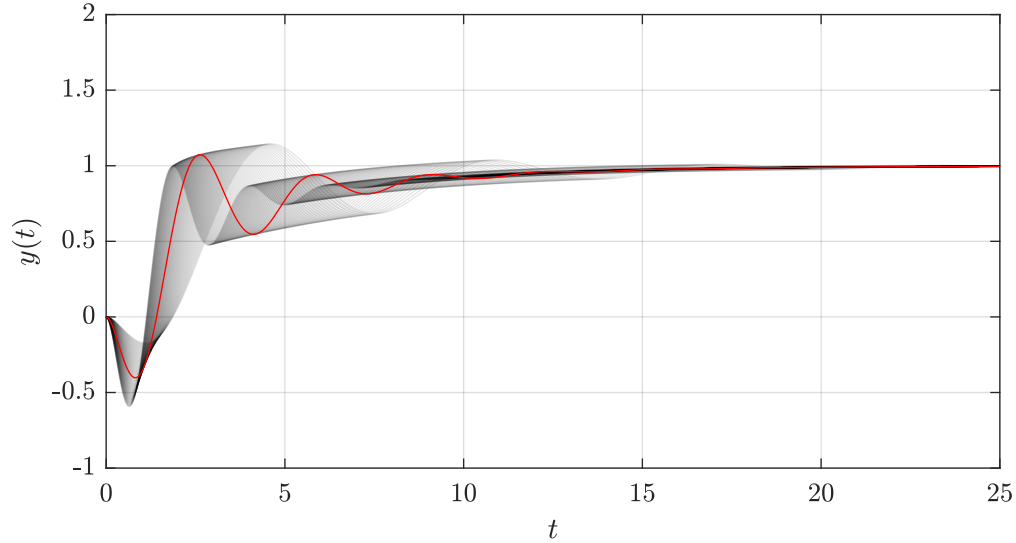
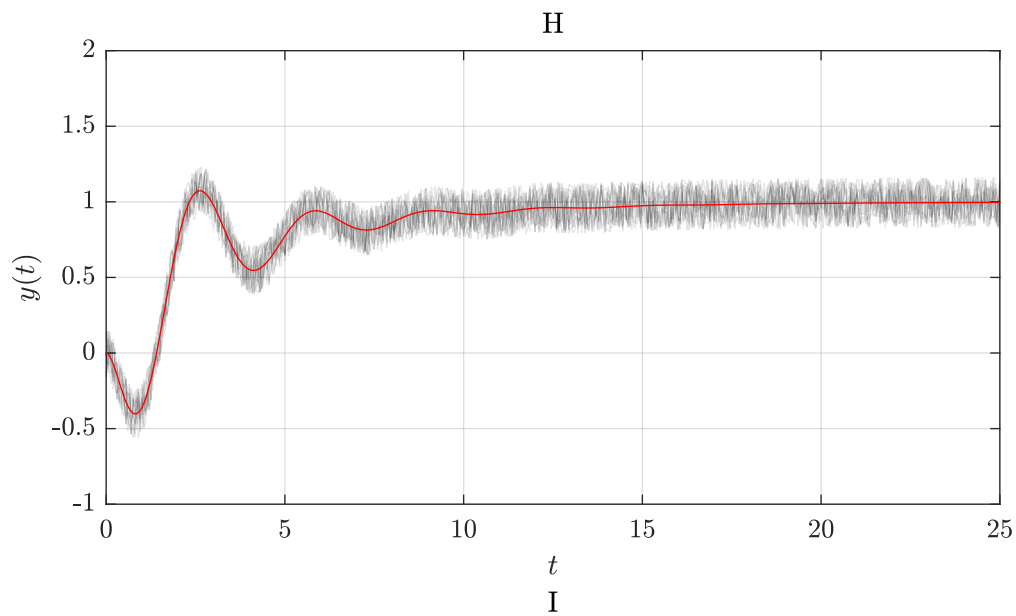
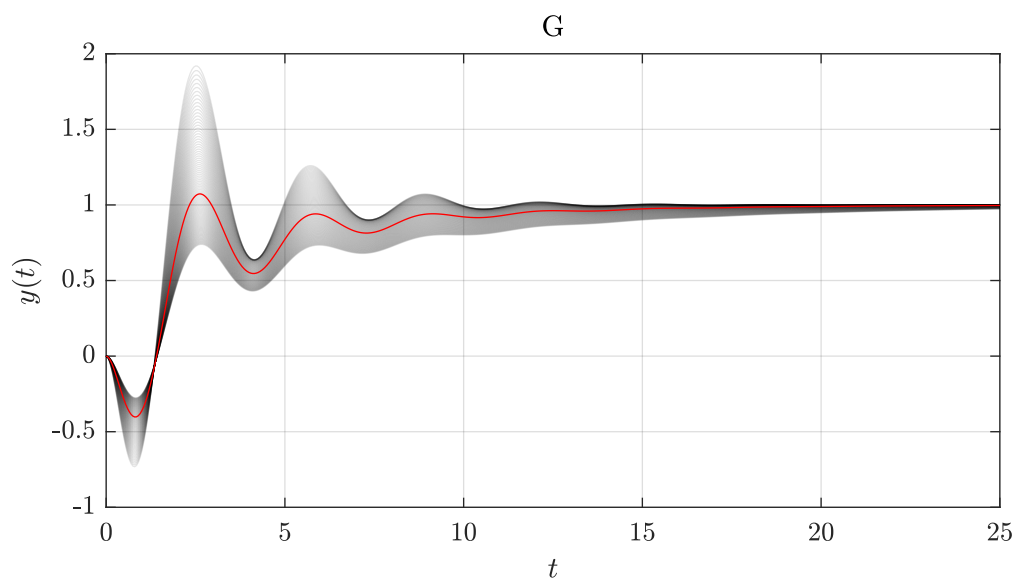


Determine the corresponding plot for each one of the following items. Provide explanations in each case.

- 1) Input noise
- 2) Output noise
- 3) $k = \hat{k} \pm 50\%$
- 4) $\tau_1 = \hat{\tau}_1 \pm 50\%$
- 5) $\tau_2 = \hat{\tau}_2 \pm 50\%$
- 6) $\zeta = \hat{\zeta} \pm 50\%$
- 7) $\omega_n = \hat{\omega}_n \pm 50\%$
- 8) $b_1 = \hat{b}_1 \pm 50\%$
- 9) $b_2 = \hat{b}_2 \pm 50\%$







MATLAB Assignments

8 Breakaway Points

Write a MATLAB function that takes a transfer function as input and,

- a) Returns the candidate breakaway points. Check your function by sketching root-locus plot of following transfer functions using *rlocus* command.

$$G_1(s) = \frac{s^2 + 2s + 2}{s(s^2 + 0.25)} , \quad G_2(s) = \frac{s^2}{(s^2 - 1)^2}$$

Hint: you might want to use *syms*, *diff*, *vpasolve* commands.

- b) How can we plot complementary root-locus using matlab functions?

- c) Manipulate your function to return the following two outputs

- 1) The points which belong to root-locus
- 2) The points which belong to complementary root-locus
- 3) The candidate points which do not belong to root-locus or its complementary plot